

## Comparative Studies of the Electrochemical and Thermal Stability of Two Types of Composite Lithium Battery Electrolytes Using Boron-Based Anion Receptors

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Comparative studies were done on two new types of boron based anion receptors, tris(pentafluorophenyl) borane (TFPB) and tris(pentafluorophenyl) borate (TFPBO), regarding conductivity enhancement, and electrochemical and thermal stability when used as additives in composite electrolytes for lithium batteries. Both additives enhance the ionic conductivity of electrolytes of simple lithium salts, LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li in several organic solvents. The electrochemical windows of TFPB based electrolytes in ethylene carbonate-propylene carbonate-dimethyl carbonate (1:1:3, v/v) are up to 5, 4.76, and 4.96 V for LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li, respectively. TFPBO has lower electrochemical stability compared to TFPB. The thermal stability of pure TFPB is better than TFPBO. Complexation with the salt improves the thermal stability of these additives. TFPB based electrolytes showed high cycling efficiencies and good cycleability when they were tested in Li/LiMn<sub>2</sub>O<sub>4</sub> cells. TFPBO based electrolytes were tested in Li/LiNiO<sub>2</sub> cells; the cells delivered high discharge capacity and good cycling efficiencies. The capacity retention of the cells using TFPB based electrolytes during multiple cycling is better than those using TFPBO based electrolytes.

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There are increasing demands for development of lithium secondary batteries with high energy density and good cyclability for energy-storage applications. Although numerous studies have been done to increase ionic conductivity and the thermal and electrochemical stability of electrolyte systems, progress in finding new stable electrolyte systems with low cost and toxicity is limited. Reduction of ion pairing in nonaqueous electrolytes is an important approach to develop high performance electrolytes for lithium secondary batteries with high energy density and good rechargeability. Usually, ion pair reduction is achieved by using solvents that have a high dielectric constant, salts that are easily dissociated, and additives that dissociate the ion pairs.<sup>1-3</sup> Cation complexing agents such as crown ethers and cryptands have been reported as additives for nonaqueous electrolytes to reduce the ion pairing and increase the conductivity.<sup>4-6</sup> Using anion receptors as additives is a more attractive approach for lithium battery electrolytes, because anion receptors enhance the dissociation of ion pairs by complexing anions. Therefore, the population of free cations in electrolytes is increased, and in turn both the conductivity and cation transference number are increased.

Recently, we reported the synthesis of a new family of boron based anion receptors as additives for lithium battery electrolytes.<sup>7</sup> They were based on the idea that an undercoordinated boron would seek a fourth ligand from the salt anion. The ability of complexing anions by boron compounds was further enhanced by fluorination of the attached groups. These boron compounds show a dramatic effect on ion pair dissociation and enhancement of solubility and conductivity of simple lithium salts, such as LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li in a 1,2-dimethoxyethane (DME) solvent. The complexation between boron compounds and anions has been proved by near edge X-ray absorption fine structure (NEXAFS) measurements.<sup>7</sup> Compared to lithium salts used for commercial lithium battery electrolytes, such as LiPF<sub>6</sub> and LiAsF<sub>6</sub>, simple lithium salts such as LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li have lower cost and are less toxic. Considering that the starting materials for the synthesis of boron compounds are also inexpensive and have low toxicity, the composite electrolytes containing boron anion receptors have the potential to be cheaply made and environmentally benign. In addition, borate compounds were found to be compatible with lithium metal in organic liquid electrolytes.<sup>8</sup> When boron anion receptors are used as additives, it is possible to replace LiPF<sub>6</sub> or LiAsF<sub>6</sub> with simple lithium salts in lithium battery electrolytes. Even LiF, which is almost insol-

uble in any organic solvent, was dissolved in DME at concentrations as high as 1 M with tris(pentafluorophenyl) borane (TFPB) as an additive.<sup>7</sup> The solution has a conductivity as high as  $6.2 \times 10^{-3}$  S/cm at 25°C. This composite electrolyte had been tested in a Li/LiMn<sub>2</sub>O<sub>4</sub> battery, and cell had been successfully cycled over 50 times.<sup>8</sup> This is the first time LiF was used as a conducting salt in a lithium battery electrolyte. However, capacity fading was a problem in that study when DME, which is electrochemically unstable, was used as the solvent.

Another boron based compound, tris(pentafluorophenyl) borate (TFPBO), also displays a comparable effect on ion pair dissociation and ionic conductivity enhancement just as TFPB.<sup>7</sup> However, high conductivity is not the only prerequisite for a lithium battery electrolyte. In this paper, electrochemical and thermal stability of composite electrolytes using boron compounds TFPB and TFPBO as additives are studied. Work on selecting the best solvent system for electrochemical stability optimization is also described, and the performance of electrolytes containing these two compounds in Li/LiMn<sub>2</sub>O<sub>4</sub> and Li/LiNiO<sub>2</sub> cells is investigated. The aim of this work is to understand better the effects of the chemical structure of boron anion receptors on their performance as additives in lithium battery electrolytes. This will provide useful information for the development of new anion receptors.

### Experimental

**Materials.**—The boron compounds TFPB and TFPBO were prepared as described in our previous paper.<sup>7</sup> Ethylene carbonate (EC) (Fluka, purity >99%), propylene carbonate (PC) (Fluka, purity >99%), dimethyl carbonate (DMC) (Aldrich, purity >99%), and 1,2-dimethoxyethane (DME) (Aldrich, purity >99%) were dried over 4 Å molecular sieves before use. The 1 M LiPF<sub>6</sub> in PC-EC-DMC (1:1:3, v/v), used as the electrolyte in the reference cell, was obtained from EM Industries, Inc. The LiMn<sub>2</sub>O<sub>4</sub> cathode material in the test cells was also purchased from EM Industries Inc. The LiNiO<sub>2</sub> cathode material was obtained from FMC.

**Conductivity measurements.**—Conductivity measurements were made at 25°C using a Hewlett-Packard 4129A impedance analyzer in the frequency range from 1 Hz to 10 MHz. Cells with Pt electrodes were used for the measurements. The cell constant ( $2.1 \text{ cm}^{-1}$ ) was determined from measurement on a 0.05 N KCl aqueous solution.

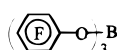
**Electrochemical stability.**—A Solartron SI 1287 electrochemical Interface System was used to perform anodic potential-sweep experiments under software control. The scan rate was 20 mV/s. The cell

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tris(pentafluorophenyl) borane (TFPB)



tris(pentafluorophenyl) borate (TFPBO)

**Figure 1.** Chemical structure of TFPB and TFPBO.

was comprised of three electrodes, a glassy carbon working electrode (7.0 mm<sup>2</sup>), a platinum wire counter electrode, and a lithium metal foil reference electrode.

**Thermal stability measurement.**—Thermogravimetric analysis (TGA) was carried out with a TA Instrumental thermal analyzer at a heating rate of 5°C/min using a sample mass of about 15 mg.

**Cell construction.**—The composite LiMn<sub>2</sub>O<sub>4</sub> and LiNiO<sub>2</sub> electrodes were made by mixing the cathode material powder with 10% acetylene black and 10% poly(vinylidene fluoride) (PVDF) (KynarFlex 2801, Atochem) in 1-methyl-2-pyrrolidinone (NMP) solvent, coating the slurry onto an aluminum foil and then drying the electrode in vacuum at 100°C. The test cell was a two-electrode hermetically sealed cell, consisting of either a LiMn<sub>2</sub>O<sub>4</sub> or LiNiO<sub>2</sub> cathode, a lithium foil anode, and a Celgard separator. Cell assembly was done in a glove box under an argon atmosphere. The lithium foil served as both the counter and reference electrodes.

**Cycling performance.**—Cycling of test batteries was carried out on an Arbin battery cycler system, and the cells were cycled at the C/3 rate under a constant current condition.

## Results and Discussion

The chemical structures of the boron anion receptors, TFPB and TFPBO, are shown in Fig. 1. The results in Table I indicate that both the borate and borane compounds yield similar conductivity enhancements for simple lithium salts. The ionic conductivities of 1 M LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li in DME solution with 1 M boron compound additives are in the range of  $5.2 \times 10^{-3}$  to  $6.8 \times 10^{-3}$  S/cm. They are much higher than the ionic conductivity for these pure salts in DME (LiF is insoluble, 1 M CF<sub>3</sub>CO<sub>2</sub>Li gives  $3.7 \times 10^{-4}$  S/cm, and 1 M C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li gives  $3.5 \times 10^{-4}$  S/cm). The ability of the boron compounds to complex anions depends on the strong electron withdrawing effect of fluorinated aryl groups attached to boron atom.<sup>7</sup> The similarity of conductivity enhancement for TFPBO and TFPB implies that the oxygen atom-bridging boron and phenyl group does not weaken the electron withdrawing effect of the fluorinated phenyl group. The relatively high molecular weight of the borate compound does not impair the conductivity of solution. The trisphenyl borate compound has a higher solubility in low-dielectric-constant organic solvents than the trisphenyl borane compound. The TFPBO dissolves in DMC in concentrations in excess of 1 M while the highest concentrations for TFPB are below 0.3 M at room temperature. This is because the bridged oxygen yields less steric hindrance from the attached groups in the TFPBO molecule than that in TFPB. TFPB has a more rigid molecular structure.

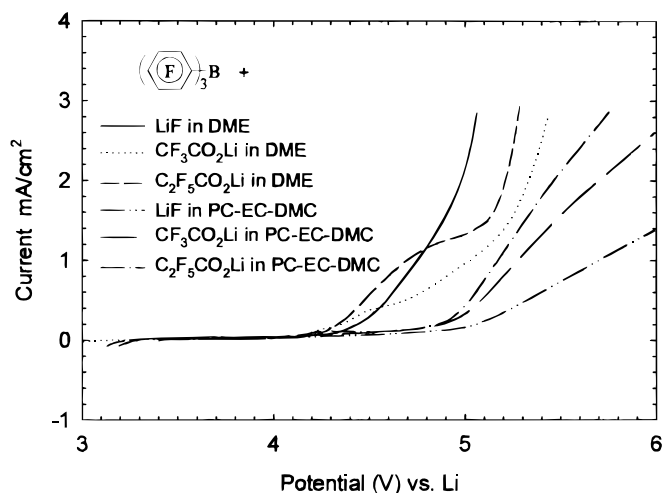
The electrochemical stability of boron anion receptors was determined by potential-sweep in electrolytes with several lithium salts. The measurements were done in solutions containing 0.2 M boron

anion receptor and 0.2 M salt. A glassy carbon electrode was used as a working electrode and a lithium foil was used as a reference electrode in this measurement. The sweep range covered the potential from the open-circuit potential (OCP) up to 5.5 V (vs. lithium) with a sweep rate of 20 mV/s. The onset potentials of anodic decomposition of composite electrolytes were determined by linear extrapolation of the increasing oxidation current. As shown in Fig. 2, the initial decompositions of the composite electrolytes in DME with TFPB as an additive occur at 4.46, 4.20, and 4.34 V for LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li, respectively. These decomposition potentials in DME are very similar to the results for LiClO<sub>4</sub> in DME.<sup>9</sup> Thus the decomposition of composite electrolytes containing TFPB in DME can be attributed to the decomposition of the solvent DME and not the additive, as supported by the fact that an electrolyte containing LiF and TFPB in DMC shows an anodic decomposition at 4.91 V.<sup>8</sup> It is well known that DME is not a suitable solvent for electrolyte for lithium batteries due to its electrochemical instability. However, this solvent was initially chosen in our studies to demonstrate the effect of the anion receptors because of its low dielectric constant. Although the electrochemical stability of the composite electrolyte containing TFPB in DMC is quite high, the solubility of TFPB in DMC is rather poor. When solvents with high dielectric constant (EC and PC) were added to DMC, the solubility of TFPB was significantly increased. Solutions in excess of 1 M TFPB in PC-EC-DMC (1:1:3, v/v) could be obtained easily. As shown in Fig. 2, the electrochemical windows of composite electrolytes with TFPB in the ternary solvent PC-EC-DMC (1:1:3, v/v) are much higher than the corresponding results in DME. The respective decomposition potentials for LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li are 5.0, 4.76, and 4.96 V. It is apparent that the electrochemical stability of the composite electrolyte is also affected by the salt component. The conductivity of these composite electrolytes in PC-EC-DMC (1:1:3, v/v) are  $3.7 \times 10^{-3}$  S/cm for CF<sub>3</sub>CO<sub>2</sub>Li,  $3.5 \times 10^{-3}$  S/cm for C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li, and  $1.2 \times 10^{-3}$  S/cm for LiF. All conductivities are lower than those found in DME (Table I). This is due to the higher viscosity of the ternary solvent PC-EC-DMC. EC and PC are usually used as the high-dielectric-constant components in electrolytes to increase ion dissociation. However, LiF is insoluble in PC-EC-DMC, and the ionic conductivities of 1 M CF<sub>3</sub>CO<sub>2</sub>Li and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li in EC-PC-DMC are  $7.1 \times 10^{-4}$  and  $8.2 \times 10^{-4}$  S/cm, which are higher than those in DME ( $3.7 \times 10^{-4}$  and  $3.5 \times 10^{-4}$  S/cm). Nevertheless, the ionic conductivities for pure salts are still lower than those of composite electrolytes with TFPB as an additive, even though the viscosities increase with the addition of the anion receptor. Therefore, the effect of boron anion receptor on ion pair dissociation is still very high even in a solvent with a very high dielectric constant.

The electrochemical stabilities of composite electrolytes with TFPBO as an additive were examined by a potential-sweep in PC-EC-DMC (1:1:3) based electrolytes. Figure 3 indicates that the respective decomposition potentials for composite electrolytes containing LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li are 4.26, 3.90, and 4.25 V. These results indicate that the electrochemical stabilities of electrolytes containing TFPBO are much lower than those containing TFPB under the same conditions. Incidentally, the decomposition

**Table I.** Ionic conductivity of solutions containing 1 M pure lithium salt or 1 M lithium salts with boron additives in DME and PC-EC-DMC (1:1:3) at 25°C.

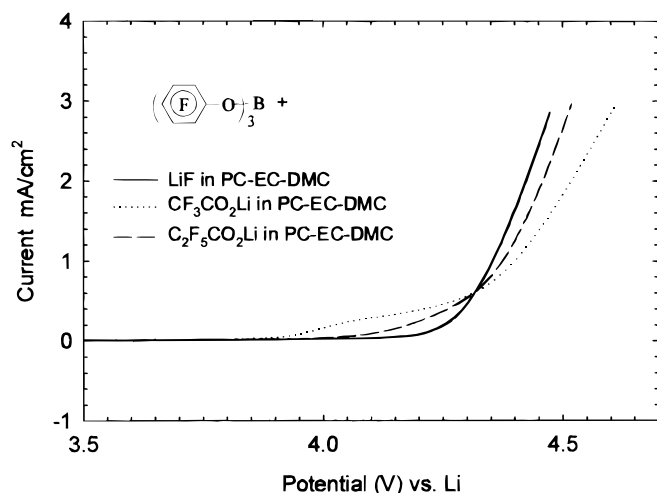
| Lithium salt                                     | Conductivity (S/cm)  |                                                 |                      |                                                  |                      |
|--------------------------------------------------|----------------------|-------------------------------------------------|----------------------|--------------------------------------------------|----------------------|
|                                                  | Pure salt            | (C <sub>6</sub> F <sub>5</sub> ) <sub>3</sub> B |                      | (C <sub>6</sub> F <sub>5</sub> O) <sub>3</sub> B |                      |
|                                                  | DME                  | DME                                             | PC-EC-DMC            | DME                                              | PC-EC-DMC            |
| LiF                                              | —                    | $6.2 \times 10^{-3}$                            | $1.2 \times 10^{-3}$ | $6.8 \times 10^{-3}$                             | $3.7 \times 10^{-3}$ |
| CF <sub>3</sub> CO <sub>2</sub> Li               | $3.7 \times 10^{-4}$ | $5.4 \times 10^{-3}$                            | $3.7 \times 10^{-3}$ | $5.2 \times 10^{-3}$                             | $3.7 \times 10^{-3}$ |
| C <sub>2</sub> F <sub>5</sub> CO <sub>2</sub> Li | $3.5 \times 10^{-4}$ | $6.0 \times 10^{-3}$                            | $3.5 \times 10^{-3}$ | $5.8 \times 10^{-3}$                             | $3.7 \times 10^{-3}$ |



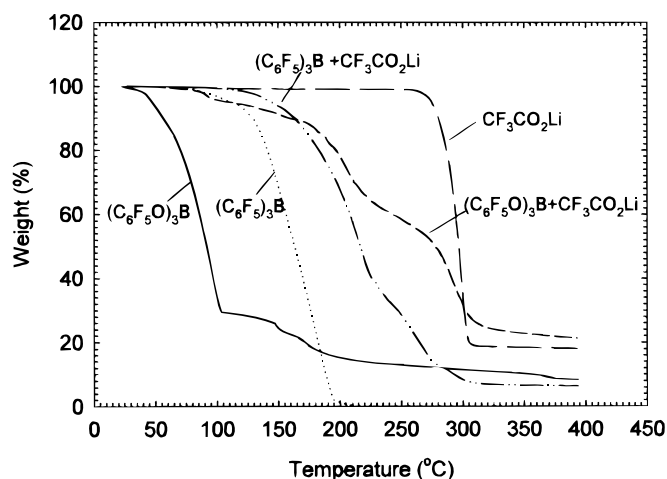
**Figure 2.** Comparison of the anodic oxidation for 0.2 M composite electrolytes with 0.2 M TFPB as an additive in DME and PC-EC-DMC (1:1:3). Glassy carbon electrode, 20 mV/s.

potential for electrolytes containing TFPBO are very close to those reported for lithium boron salts with similar chemical structures.<sup>10</sup> For instance, fluorinated lithium bis[1,2-benzenedio-lato(2-)-O,O']borate was found to decompose at 4.1 V in PC. The difference in electrochemical stability of the aryl borane and aryl borate compounds is most likely due to the electrochemical oxidation of these two boron compounds being initiated by the formation of a radical. This oxidation mechanism has been proposed to describe the decomposition process of bisarylenediolateborate lithium salts.<sup>11</sup> Due to an oxygen atom-bridging phenyl group and boron atom, one of the lone-pair electrons in oxygen is easy to lose, and a radical is formed to initiate its decomposition. However, the fluorination of the phenyl group in TFPBO has decreased the electron density on the oxygen atom and increased the electrochemical stability of this type of compound.<sup>11</sup> It is apparently the bridged oxygen that makes TFPBO have less electrochemical stability and a lower electrochemical decomposition potential than TFPB.

The thermal stability of TFPB and TFPBO borate was examined by thermogravimetric analysis in air. As shown in Fig. 4, the initial weight loss of TFPB occurs at 79°C, while TFPBO starts losing weight from the beginning of heating and has a major weight loss at 42°C. Because sublimation of TFPB and TFPBO only occurs at high temperature and low pressure (100 to 110°C at 20 Pa), these weight



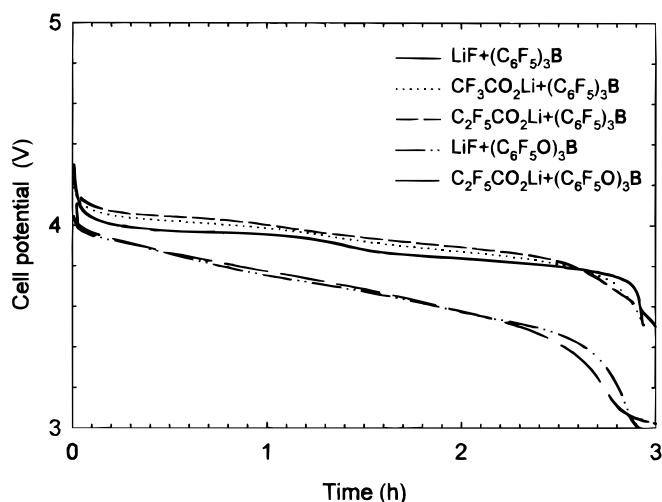
**Figure 3.** Comparison of the anodic oxidation for 0.2 M composite electrolytes with 0.2 M TFPBO as an additive in PC-EC-DMC (1:1:3). Glassy carbon electrode, 20 mV/s.



**Figure 4.** The thermogravimetric analysis curves for TFPB, TFPBO, and their complexes with lithium trifluoroacetate (1:1).

losses in their TGA curves are considered mainly from the thermal decomposition in air. Their thermal stabilities were also measured in the presence of lithium salt. The mixture of the boron compound and  $\text{CF}_3\text{CO}_2\text{Li}$  was obtained by evaporation of a DME solution containing stoichiometric amounts (1:1) of boron compound and salt.  $\text{CF}_3\text{CO}_2\text{Li}$  is thermally stable, and its decomposition occurs at 265°C. The TGA curves of these mixtures show decomposition at 117°C for the TFPB and 77°C for TFPBO. The respective decomposition temperatures are higher than those of pure boron compounds only, but lower than pure  $\text{CF}_3\text{CO}_2\text{Li}$  salt. Therefore it is reasonable to assume that the mixture which is obtained from the solution containing the boron compound and the lithium salt is not a simple physical mixture. The different decomposition temperature is an indication of strong complexation between the boron compound and the lithium salt. This result is consistent with the results from NEX-AFS.<sup>7</sup> Thus, the complexation between the boron anion receptor and the lithium salt increases the thermal stability of the boron anion receptor. The thermal stability for TFPBO and its complex with the lithium salt are much lower than those for TFPB. This implies that a phenyl-O-B bond is not only less resistant to electrochemical oxidation but also more susceptible to thermal oxidation than the phenyl-B bond.

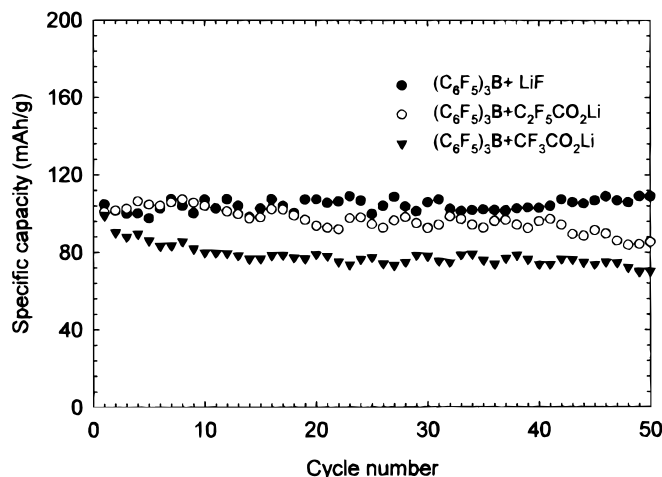
Having established high conductivity and good electrochemical and thermal stability of these composite electrolytes with TFPB and TFPBO as additives, the feasibility of using these new composite electrolytes was tested in lithium cells. In our recent report,<sup>8</sup> a Li/LiMn<sub>2</sub>O<sub>4</sub> battery with electrolyte containing TFPB and LiF in DME was already cycled over 50 times.<sup>8</sup> The present results confirm that severe capacity fading was observed due to the instability of the DME solvent. However, the present results indicate that electrolytes containing TFPB and lithium salts in PC-EC-DMC have excellent electrochemically stability. Therefore, they were tested in Li/LiMn<sub>2</sub>O<sub>4</sub> cells. The Li/LiMn<sub>2</sub>O<sub>4</sub> cells were cycled between 3.5 and 4.4 V at the C/3 rate. Figure 5 shows the discharge curves for Li/LiMn<sub>2</sub>O<sub>4</sub> cells at 25°C at the C/3 rate with 1 M solutions of LiF,  $\text{CF}_3\text{CO}_2\text{Li}$ , and  $\text{C}_2\text{F}_5\text{CO}_2\text{Li}$  containing 1 M TFPB as an additive in PC-EC-DMC (1:1:3, v/v). The two plateaus around 4 V for each discharge curve is the indication of  $\text{Li}^+$  intercalation into spinel manganese oxide. The plateau for LiF solution is lower than that of  $\text{CF}_3\text{CO}_2\text{Li}$  or  $\text{C}_2\text{F}_5\text{CO}_2\text{Li}$ . This is because of the lower ionic conductivity of the LiF electrolyte. The initial discharge capacities of cells using electrolyte containing LiF,  $\text{CF}_3\text{CO}_2\text{Li}$ , and  $\text{C}_2\text{F}_5\text{CO}_2\text{Li}$  salts were 105 mAh g<sup>-1</sup>, 101 mAh g<sup>-1</sup>, and 99 mAh g<sup>-1</sup>, respectively. They are very close to the capacity (102 mAh g<sup>-1</sup>) obtained with a reference cell using commercially available electrolyte [LiPF<sub>6</sub> in PC-EC-DMC (1:1:3, v/v)]. The discharge plots (capacity vs. cycle number) are shown in Fig. 6. The electrolyte containing LiF and



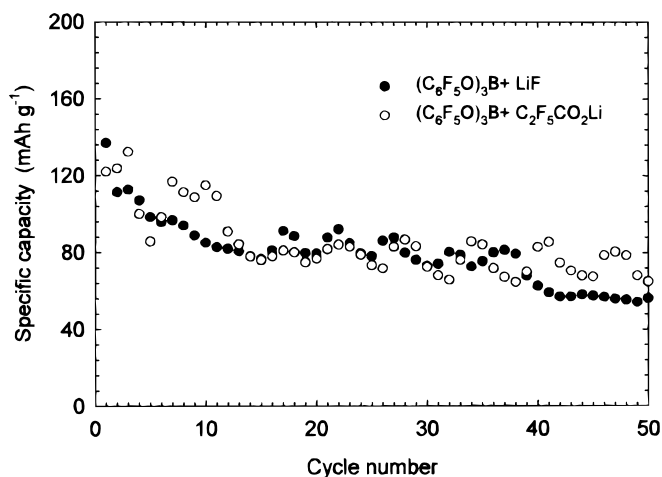
**Figure 5.** Discharge curves for Li/LiMn<sub>2</sub>O<sub>4</sub> cells with composite electrolytes containing 1 M TFPB as additive and Li/NiO<sub>2</sub> cells with composite electrolytes containing 1 M TFPBO as additive in PC-EC-DMC (1:1:3, v/v) at 25°C at C/3 rate.

TFPB has the best capacity maintenance. No capacity fading was observed over 50 cycles. These results are much superior to those obtained with an electrolyte containing LiF and TFPB in DME (59 mAh g<sup>-1</sup> in 51st discharge).<sup>8</sup> The electrolytes with the CF<sub>3</sub>CO<sub>2</sub>Li and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li salts showed less stability during cycling compared to the LiF salt. The respective capacities for the 50th cycle were 71 and 86 mAh g<sup>-1</sup>. These results are consistent with the electrochemical stability measurements. The coulombic discharge/charge efficiencies in all these electrolytes are about 82% in the first cycle and 94-100% during the subsequent cycles. These results are pretty close to the efficiencies obtained from a Li/LiMn<sub>2</sub>O<sub>4</sub> reference cell with commercial electrolyte LiPF<sub>6</sub> in PC-EC-DMC (83% in first cycle, 93-100% in subsequent cycles) in the same cycling condition. Considering that the difference between charge and discharge capacity is drastically reduced for the subsequent cycling, the low coulombic efficiency in the first cycle can be due to the activation of the cathode material.

The electrochemical window for TFPB is high enough for some cathode materials such as V<sub>2</sub>O<sub>5</sub> and LiNiO<sub>2</sub>. LiNiO<sub>2</sub> has been studied extensively recently, and its analogues have been considered as attractive substitutes for LiCoO<sub>2</sub>, which is presently used in commercial lithium batteries. The operating voltage for LiNiO<sub>2</sub> vs. lithi-



**Figure 6.** Specific capacity vs. cycle number for Li/LiMn<sub>2</sub>O<sub>4</sub> cells using composite electrolytes 1 M LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li with 1 M TFPB as an additive in PC-EC-DMC (1:1:3, v/v) at 25°C at C/3 rate.



**Figure 7.** Specific capacity vs. cycle number for Li/LiNiO<sub>2</sub> cells using composite electrolytes 1 M LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li with 1 M TFPB as an additive in PC-EC-DMC (1:1:3, v/v) at 25°C at C/3 rate.

um is 3.7 V. Composite electrolytes containing LiF and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li with TFPB as additive in PC-EC-DMC (1:1:3, v/v) were tested in Li/LiNiO<sub>2</sub> cells. The test cells for Li/LiNiO<sub>2</sub> had the same construction as Li/LiMn<sub>2</sub>O<sub>4</sub> cell. They were cycled between 2.5 and 4.15 V at 25°C at the C/3 rate. The respective discharge capacities with LiF and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li electrolytes were 136 and 122 mAh g<sup>-1</sup> in the first cycle. They are higher than the discharge capacity (105 mAh g<sup>-1</sup> in the first cycle) when LiPF<sub>6</sub> in PC-EC-DMC (1:1:3, v/v) was used as the electrolyte under the same conditions. These cells were cycled over 50 times. Their discharge capacities vs. cycle number are shown in Fig. 7 and indicate capacity loss with cycling. The coulombic charge/discharge efficiencies of the batteries were about 40% in the first cycle and exceed 84% for the subsequent cycles. The cycling performance of composite electrolytes with TFPBO as additive was not as good as for the electrolytes using TFPB. The difference in capacity maintenance during cycling in cells using TFPB and TFPBO as additives in electrolytes is clearly related to the electrochemical stability of these compounds.

## Conclusions

Fluorinated boron based anion receptors, TFPB and TFPBO, were studied as additives for composite electrolytes for lithium batteries. Although both additives have a high molecular weight, they increase the solubility and conductivity of electrolytes with simple lithium salts such as LiF, CF<sub>3</sub>CO<sub>2</sub>Li, and C<sub>2</sub>F<sub>5</sub>CO<sub>2</sub>Li, even in solvents with high viscosity (PC-EC-DMC). TFPB has good electrochemical stability in PC-EC-DMC. The electrochemical stability of composite electrolytes is also affected by the salt component. With LiF, the electrochemical window of the electrolyte is as high as 5.0 V. The electrolytes with TFPBO as an additive are less stable under the same conditions. TFPB is thermally stable in air to 77°C, and TFPBO decomposed at 42°C. However, complex formation with the lithium salt significantly improves their thermal stability in air. Composite electrolytes with TFPB as an additive were tested in a 4 V Li/LiMn<sub>2</sub>O<sub>4</sub> cell. They displayed excellent cycling performance particularly in the electrolyte using LiF. Almost no apparent capacity fade was observed over 50 cycles, and the cell cycled with a high charge/discharge efficiency (above 94%). TFPBO was also tested in a Li/LiNiO<sub>2</sub> cell. The cell had a higher initial discharge capacity than cells with an electrolyte of LiPF<sub>6</sub> in PC-EC-DMC. For Li/LiNiO<sub>2</sub> cells, the capacity fades with cycling. This is attributed to the lower stability of the borate additive.

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